



This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

SPRING AI

GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

Simone Scannapieco

Corso avanzato per Venis S.p.A, Venezia, Italia

Novembre 2025

Note

[illegible]



TEXT-TO-SPEECH

Note

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

- Simone Scannapieco

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the paper.

```
public interface TextToSpeechModel extends Model<TextToSpeechPrompt, TextToSpeechResponse>,
    StreamingTextToSpeechModel {

    default byte[] call(String text) {
        TextToSpeechPrompt prompt = new TextToSpeechPrompt(text);
        ModelResult<byte[]> result = call(prompt).getResult();
        if (result == null) {
            return new byte[0];
        }
        byte[] output = result.getOutput();
        return (output != null) ? output : new byte[0];
    }

    @Override
    TextToSpeechResponse call(TextToSpeechPrompt prompt);

    default TextToSpeechOptions getDefaultOptions() {
        return TextToSpeechOptions.builder().build();
    }
}
```

-  Simone Scannapieco

Note

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the paper.

```
@RestController
@RequestMapping("/api")
public class AudioController {

    private final TextToSpeechModel speechModel;

    AudioController(TextToSpeechModel speechModel) {
        this.speechModel = speechModel;
    }

    @GetMapping("/speech")
    String speech(@RequestParam("message") String message) throws IOException {
        byte[] audioBytes = speechModel.call(message);
        Path path = Paths.get("output.mp3");
        Files.write(path, audioBytes);
        return "MP3 salvato con successo in " + path.toAbsolutePath();
    }
}
```

-  Simone Scannapieco

Note

[illegible]

Controller di sintetizzazione audio

```

@RestController
@RequestMapping("/api")
public class AudioController {
    ...

    @GetMapping("/speech-options")
    String speechWithOptions(@RequestParam("message") String message) throws IOException {
        TextToSpeechResponse speechResponse = speechModel.call(
            new TextToSpeechPrompt(message,
                OpenAiAudioSpeechOptions.builder()
                    .voice(OpenAiAudioApi.SpeechRequest.Voice.NOVA)
                    .speed(2.0f)
                    .responseFormat(OpenAiAudioApi.SpeechRequest.AudioResponseFormat.MP3)
                    .build());
            Path path = Paths.get("speech-options.mp3");
            Files.write(path, speechResponse.getResult().getOutput());
            return "MP3 salvato con successo in " + path.toAbsolutePath();
        }
    }
}

```

- ➔ Popolato un `TextToSpeechPrompt` con tutte le sovrascritture volute rispetto al *default*
- ➔ `speechResponse.getResult().getOutput()` ripulisce l'output da tutti i metadati restituendo il dato grezzo (`byte[]`)

Note

[illegible]

- ## Approccio REST per sintetizzazione vocale: Gemini - singolo

7/17

Note

[illegible]

```
curl "https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash-preview-tts:generateContent" \
-H "x-goog-api-key: $GEMINI_API_KEY" -X POST -H "Content-Type: application/json" \
-d '{
  "contents": [{
    "parts": [{
      "text": "Mario: Come va oggi Maria? Maria: Abbastanza bene, te?"
    }]
  }],
  "generationConfig": {
    "responseModalities": ["AUDIO"],
    "speechConfig": {
      "multiSpeakerVoiceConfig": {
        "speakerVoiceConfigs": [{
          "speaker": "Joe",
          "voiceConfig": {
            "prebuiltVoiceConfig": {
              "voiceName": "Kore"
            }
          }
        }, {
          "speaker": "Jane",
          "voiceConfig": {
            "prebuiltVoiceConfig": {
              "voiceName": "Puck"
            }
          }
        }
      ]
    }
  }
},
  "model": "gemini-2.5-flash-preview-tts",
}' | jq -r '.candidates[0].content.parts[0].inlineData.data' | base64 --decode > out.pcm

ffmpeg -f s16le -ar 24000 -ac 1 -i out.pcm out.wav
```

Note

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

➔ Servizio *text-to-speech* Gemini

- 1 Modifica dipendenze di progetto
- 2 Creazione modelli Gemini TTS *request* e *response*
- 3 Creazione modello per informazioni voci disponibili
- 4 Creazione convertitore *output* audio Gemini a formato riproducibile nel *web*
- 5 Modifica controllore MVC
- 6 Test delle funzionalità con Postman/Insomnia

Note

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the paper.

Dipendenze applicativo

```

...
<dependency>
  <groupId>com.google.cloud</groupId>
  <artifactId>google-cloud-texttospeech</artifactId>
  <version>2.40.0</version>
</dependency>
...

```

Note

Note area with horizontal lines for writing.

```
package it.venis.ai.spring.demo.model;

public class GeminiTtsRequest {
    private String text;
    private String voice;
    private String model;
    private String stylePrompt;

    ...
}
```

[illegible]

```
package it.venis.ai.spring.demo.model;

public class GeminiTtsResponse {
    private String audioBase64;
    private String mimeType;
    private String message;

    public GeminiTtsResponse(String audioBase64, String mimeType, String message) {
        this.audioBase64 = audioBase64;
        this.mimeType = mimeType;
        this.message = message;
    }

    ...
}
```

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

```
package it.venis.ai.spring.demo.util;

...

public class PcmToWavConverter {

    public static byte[] pcmToWav(byte[] pcmData, float sampleRate, int sampleSizeInBits,
                                   int channels, boolean signed, boolean bigEndian) throws IOException {

        // Crea la specifica del formato audio
        AudioFormat audioFormat = new AudioFormat(
            sampleRate,           // Frequenza di campionamento (Hz)
            sampleSizeInBits,     // Dimensione del campione in bit
            channels,             // Numero di canali
            signed,               // Con segno o senza segno
            bigEndian             // Big-endian o little-endian
        );

        // Racchiude i dati PCM in un ByteArrayInputStream
        ByteArrayInputStream pcmInputStream = new ByteArrayInputStream(pcmData);

        // Calcola la lunghezza del frame
        long frameLength = pcmData.length / audioFormat.getFrameSize();

        // Crea un AudioInputStream dai dati PCM
        AudioInputStream audioInputStream = new AudioInputStream(
            pcmInputStream,
            audioFormat,
            frameLength
        );

        ...
    }
}
```

Note

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

```
package it.venis.ai.spring.demo.model;

import java.util.List;

public class VoiceInfo {
    private String name;
    private List<String> languageCodes;
    private String gender;

    public VoiceInfo(String name, List<String> languageCodes, String gender) {
        this.name = name;
        this.languageCodes = languageCodes;
        this.gender = gender;
    }

    ...
}
```

[illegible]

```
package it.venis.ai.spring.demo.controllers;
...
@RestController
@RequestMapping("/gemini/multi-modality")
public class GeminiMultimodalityController {

    /** Chiave API di Google AI recuperata dalle proprietà di configurazione */
    @Value("${GOOGLE_AI_API_KEY}")
    private String geminiApiKey;

    /** URL base dell'API Gemini per le richieste ai modelli */
    @Value("${gemini.api.models.url}")
    private String geminiApiUrl;

    /** Servizio per le operazioni multimodali (STT, ITT, TTS) */
    private final MultiModalityService multiModalityService;

    /** Client HTTP REST per le chiamate all'API di Gemini */
    private final RestTemplate restTemplate = new RestTemplate();

    /**
     * Costruttore del controller per la gestione delle funzionalità multimodali di Gemini.
     *
     * @param multiModalityService servizio per le operazioni multimodali (STT, ITT, TTS)
     */
    public GeminiMultimodalityController(MultiModalityService multiModalityService) {

        this.multiModalityService = multiModalityService;
    }
    ...
}
```

Note

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

```
<!DOCTYPE html>
<html lang="it">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Gemini Text-to-Speech Demo</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }

    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
      min-height: 100vh;
      padding: 20px;
      display: flex;
      justify-content: center;
      align-items: center;
    }

    .container {
      background: white;
      border-radius: 20px;
      box-shadow: 0 20px 60px rgba(0,0,0,0.3);
      padding: 40px;
      max-width: 900px;
      width: 100%;
    }

    ...
  </body>
</html>
```

Note

[illegible]

17 / 17

[illegible]